温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

Descriptive Statistics

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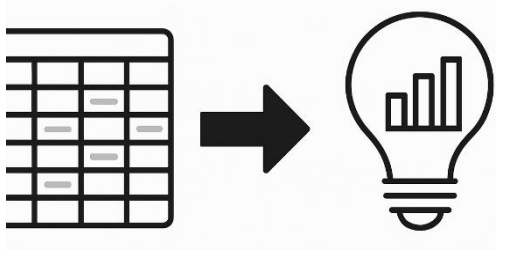
Today’s Agenda

- Opening
- Population vs. Sample
- Measures of Location
  - Mean, median, mode – different ways to identify “the center”
- Measures of Dispersion
  - Range, percentile, variance, standard deviation – understanding variability
- Measures of Shape
  - Skewness, kurtosis – symmetry and peakedness of data
- Outliers
  - Identification & treatment

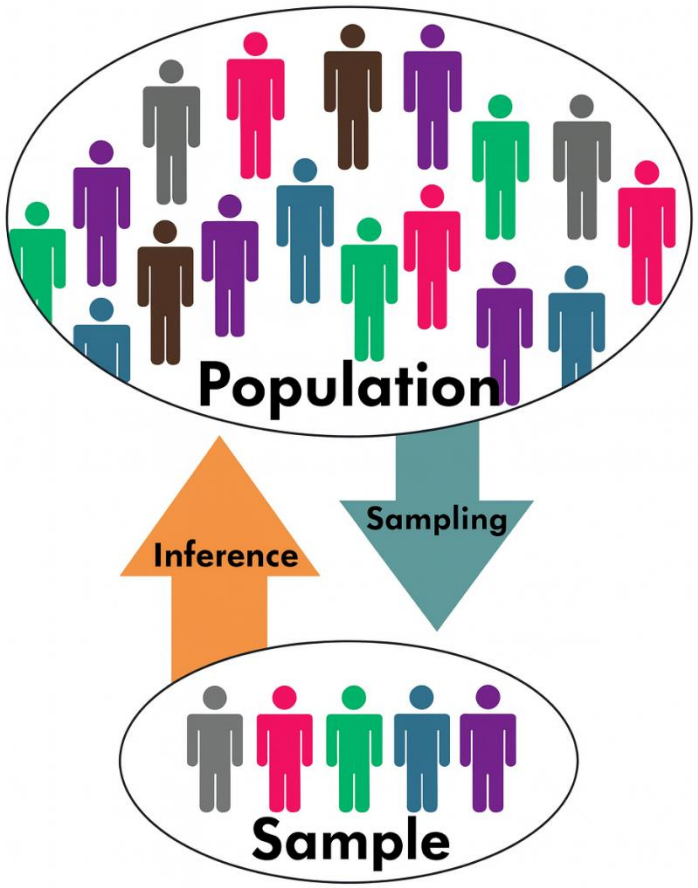
Opening – Why Descriptive Statistics?

- When you first open a dataset, what do you look at first?
  - Size of dataset
  - Number of rows (observations) / columns (variables)
  - Variables definitions / types
  - Missing values
  - Unusual numbers
  - …



- How do we **summarize** and **make sense** of all the numbers?
- This is where Descriptive Statistics comes in
  - Provides tools to **condense large datasets** into meaningful summaries
  - Helps us spot **patterns**, identify **outliers**, and understand **variability**
  - Builds the **foundation** for deeper analysis (predictive & prescriptive analytics)

Population vs. Sample



Population

- Entire group of interest for *a specific decision or investigation*
- *Example:* All WKU students

Sample

- A subset of the population selected for analysis
- *Example:* Students in MGS 3101 W03, Fall 2025

Goal

- Sampling provides sufficient information to make valid inferences about the population

Measures of Location – Mean

- Definition
  - Arithmetic average: the sum of all values divided by the number of observations
  - "center of gravity", "balance point"
- Formulas
  - For a population of size N:

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$$

  - For a sample of n observations:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

N: total population size
n: sample size
x_i: the ith observation

- Key Points
  - Uses all data values → **sensitive** to extreme values (**outliners**)
  - Often used in business to describe average cost, revenue, time, or performance

Measures of Location – Median

- Definition
  - The **middle** value of data when data are arranged in **ascending order**
  - It divides the data into two equal halves
- Rules
  - If n is **odd**: the median is the middle observation
  - If n is **even**: the median is the average of the two middle observations
- Why Median?
  - Less affected by outliners and skewed data compared to the mean
  - Useful for income, housing prices, or skewed distributions

Measures of Location – Mode

- Definition
  - Values that occurs **most frequently** in data
  - For grouped data, it is the class interval (group) with the **highest frequency**
- Rules

  A dataset may have:
  - One mode → *Unimodal*
  - Two modes → *Bimodal*
  - More than two modes → *Mutimodal*
- Why Mode?
  - The only measure of central tendency suitable for **categorical data** (e.g., most popular brand, favorite color)
  - Useful in identifying the **most common outcome** in a dataset

Comparing Mean, Median, and Mode

- Example Dataset (Annual Income in $1,000s)

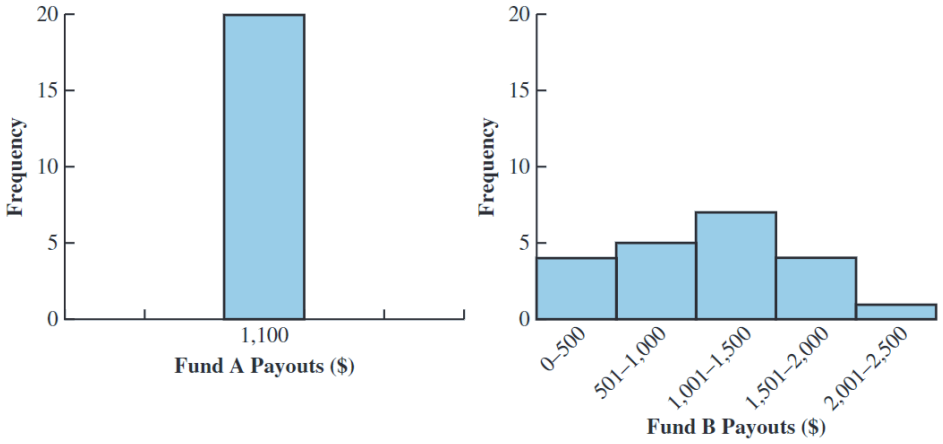
  25, 28, 30, 32, 35, 36, 40, 42, 120

  - Mean = 43.1
  - Median = 35
  - Mode = none (no value repeats)
- Interpretation
  - Mean (43.1) is pulled upward by the extreme income of 120 → not very representative
  - Median (35) better represents the "typical" income in this group
  - Mode is not helpful here because all values are unique
- When to Use Each?
  - Mean: symmetric data without outliers (e.g., test scores)
  - Median: skewed data or with outliers (e.g., income, house prices)
  - Mode: categorical or discrete data (e.g., favorite product, most frequent defect type)

Measures of Dispersion

- **Fund A:** mean payout = $1,100, but no variability
- **Fund B:** same mean payout, but payouts vary widely

→ Fund B is riskier due to greater dispersion



From Camm Ch2

- Dispersion describes the spread or variability of data values
- It tells us how tightly or loosely data points cluster around the mean
- Key measures: range, percentile, variance, standard deviation (SD)

Measures of Dispersion – Range

- Definition
  - Difference between the largest and smallest values in a dataset
  - The simplest measure of dispersion
- Formula

$$Range = Maximum\,Value\ - Minimum\,Value$$

- Why Range?
  - Quick to easy to calculate
  - Provides a rough idea of data spread
  - Limitation: Only depends on two values (mix & max), so it can be heavily influenced by **outliners**.

Measure of Dispersion – Percentiles and Quartiles

- Percentiles
  - To find the *p-th* percentile, first sort data in ascending order, the *p-th* percentile is the value below which $p\%$ of the data fall
- Quartiles (special percentiles)
  - Quartiles split an ordered dataset into four equal parts
    - Q1 (First Quartile): 25th percentile, value below which 25% of the data fall
    - Q2 (Second Quartile / Median): 50th percentile, middle value of the dataset
    - Q3 (Third Quartile): 75th percentile, value below which 75% of the data fall
- Interquartile Range (IQR)
  - The spread of the middle 50% of the data
  - Formula: $IQR = Q_3 - Q_1$
- Interpretation
  - A measure of variability that is not influenced by extreme values (unlike the range)
  - Useful for detecting outliers

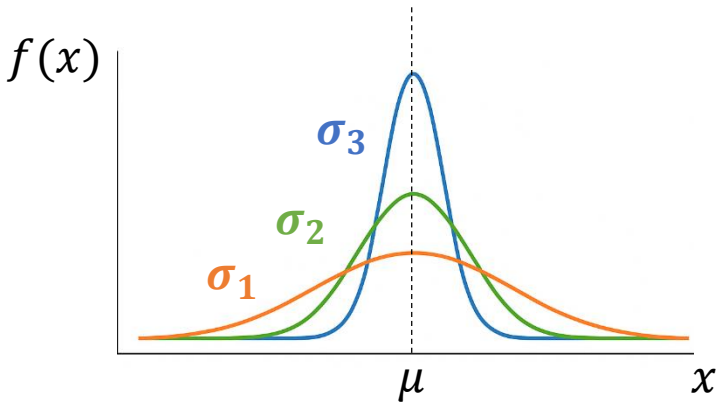
Measures of Dispersion – Variance

- Definition
  - Variance measures how far data points **deviate from mean**, **on average**
  - It is the average of the squared deviations from the mean
- Formulas
  - For a population: $\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$
  - For a sample: $s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$

- ❑ Larger variance = more spread in the data
- ❑ Units are **squared**, so interpretation is less intuitive → motivates the standard deviation

- Why Two Formulas?
  - In practice, we usually work with samples, not entire populations
  - Dividing by $\boldsymbol{n-1}$ (instead of $\boldsymbol{n}$) gives an **unbiased** estimate of the population variance
  - This correction is called **Bessel's correction**

Measures of Dispersion – Standard Deviation (SD)



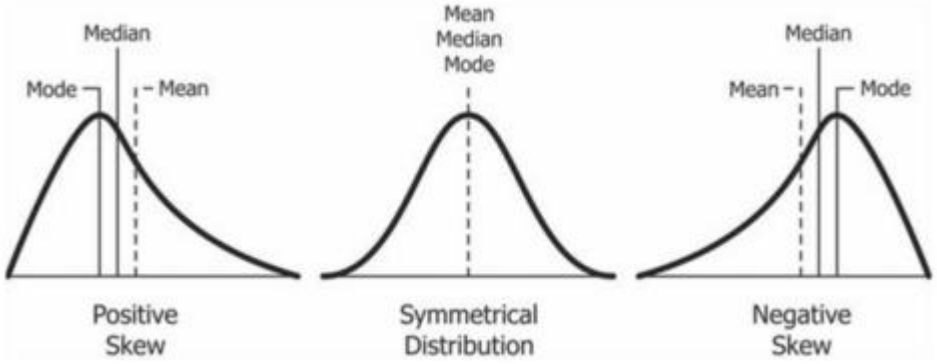
- Definition
  - The positive **square root** of the variance
  - Measure the average distance of data values form the mean
- Formula

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}} \qquad s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

- Interpretation
  - Unlike variance, the SD is expressed in the **same units as the data**, making it easier to interpret
  - Small SD → data points are close to the mean (low variability)
  - Large SD → data points are widely spread (high variability)
  - In finance, SD is often used as a **measure of risk**

Measures of Shape – Skewness

- Definition
  - Skewness describes the **asymmetry** of a data distribution relative to the **mean**.
  - A perfectly symmetric distribution (like a normal curve) has skewness = 0.
- Types of Skewness
  - Positively Skewed (Right-Skewed)
    - Right tail is longer
    - Mean > Median > Mode
    - Example: income distribution
  - Negatively Skewed (Left-Skewed)
    - Left tail is longer
    - Mean < Median < Mode
    - Example: exam scores when most student score high



https://en.wikipedia.org/wiki/Skewness

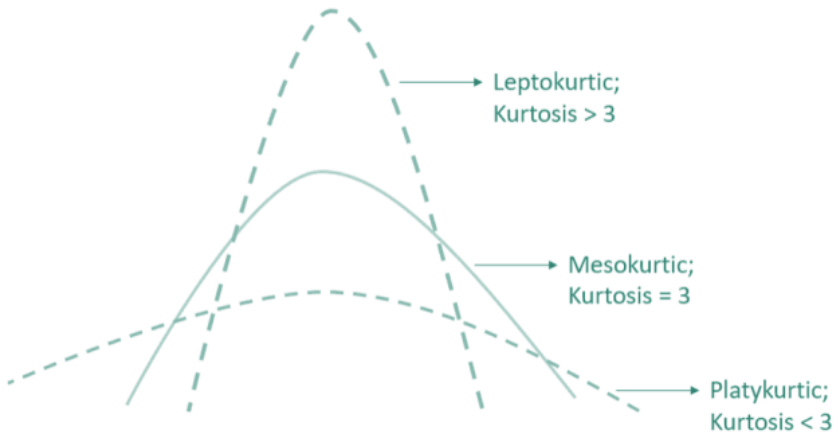
Measures of Shape – Skewness

- Coefficient of Skewness (CS)
  - $CS < 0$: left-skewed
  - $CS > 0$: right-skewed
  - Magnitude indicates degree of skewness
    - $|CS| > 1$: high skewness
    - $0.5 \leq |CS| \leq 1$: moderate skewness
    - $|CS| < 0.5$: Approx. symmetric

$$CS = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^3}{\sigma^3}$$

- Why It Matters?
  - Skewness shows whether the **mean** or **median** better represents the data.
  - Important in finance (returns often skewed), economics (income, wealth), and quality control.

Measures of Shape – Kurtosis

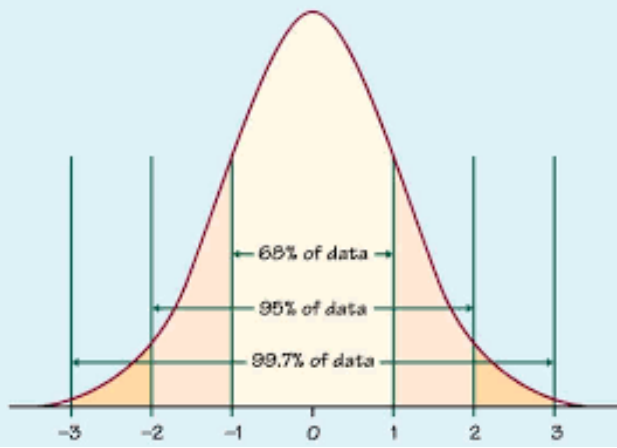


- Definition
  - Kurtosis measures the shape of a distribution, focusing on its **peakedness** and **tail heaviness** compared to a normal distribution.
  - It shows whether data are concentrated near the mean or spread more in the tails.
- Coefficient of Kurtosis (CK)

$$CK = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^4}{\sigma^4}$$

- Types of Kurtosis
  - Mesokurtic ($CK \approx 3$): normal distribution
  - Leptokurtic ($CK > 3$): tall, narrow peak, heavy tails; more outliners and extreme values
  - Platykurtic ($CK < 3$): flatter, wider peak, light tails; fewer outliners and extreme values

Standardized Values (z-score)

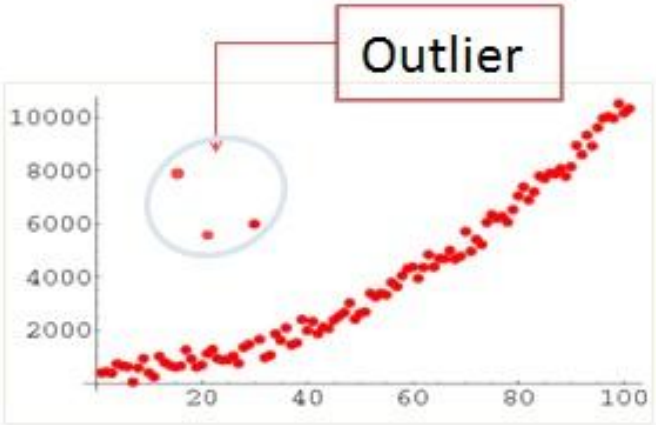


- Definition
  - A z-score is a standardized value that measures how many standard deviations an observation is from the mean.
  - It allows comparisons across different datasets and units.
- Formula

$$z_i = \frac{x_i - \mu}{\sigma} \qquad z_i = \frac{x_i - \bar{x}}{s}$$

- Interpretation
  - $z = 0$: observation is exactly at the mean
  - $z = +2$: observation is 2 SD above the mean
  - $z = -1.5$: observation is 1.5 SD of the mean
- Empirical Rule (for normal distributions)
  - $\sim 68\%$ of values fall within ± 1 SD of the mean
  - $\sim 95\%$ within ± 2 SD
  - $\sim 99.7\%$ within ± 3 SD

Outliners



- Definition
  - **Unusually small** or **large** values
  - No single universal definition, but statistical rules help us identify them
- Identification (Rules of Thumb)
  - z-scores rule: If $|z| > 3$, the value may be an outliner
  - Interquartile Range (IQR) rule:
    - Mild outliners: $1.5 \times IQR$ beyond Q_1 or Q_3
    - Extreme outliner: $3 \times IQR$ beyond Q_1 or Q_3
- Possible Causes
  - Data entry error
  - Incorrectly recorded or included observation
  - Genuine extreme value (important to the analysis)
- Handling Outliners
  - **Keep** them if they represent valid but unusual cases
  - **Remove** them if they are clear mistakes or irrelevant
  - **Correct** them if caused by obvious error in recording

Reflection

- What we learned today
  - Population vs. Sample → the basis for statistical inference
  - Measures of Location (mean, median, mode) → different ways to describe the "center"
  - Measures of Dispersion (range, percentiles, variance, standard deviation, IQR) → describe variability and consistency
  - Measures of Shape (skewness, kurtosis) → reveal asymmetry and peakedness
  - Outliers → how to identify them and decide whether to keep, remove, or correct
- Why it matters
  - Helps us **spot patterns**, **detect unusual values**, and **understand variability**
  - Forms the **foundation** for deeper analysis (predictive and inferential statistics)
  - Reminds us that the choice of statistic depends on the type of data
    - *Categorical (nominal)* → mode, proportions
    - *Ordinal* → median, percentiles
    - *Interval/Ratio* → mean, variance, standard deviation
- Looking ahead
  - Next class: **Measures of Association** (covariance, correlation) and **hands-on practice in Python**